Usage-Aware Quantization and Horizon-Weighted Prediction for Enhanced RPQ-Based Speech Self-Supervised Learning

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INTRODUCTION

Random-projection-quantizer-based speech self-supervised learning (RPQ-SSL) generates discrete pseudo-labels online. Two limitations remain: **imbalanced codebook utilization** and **uniform loss weighting** across future prediction horizons.

Our approach: UA-RPQ improves pseudo-label generation. HW-NTP accounts for differences in future-token prediction difficulty.

PROPOSED FRAMEWORK

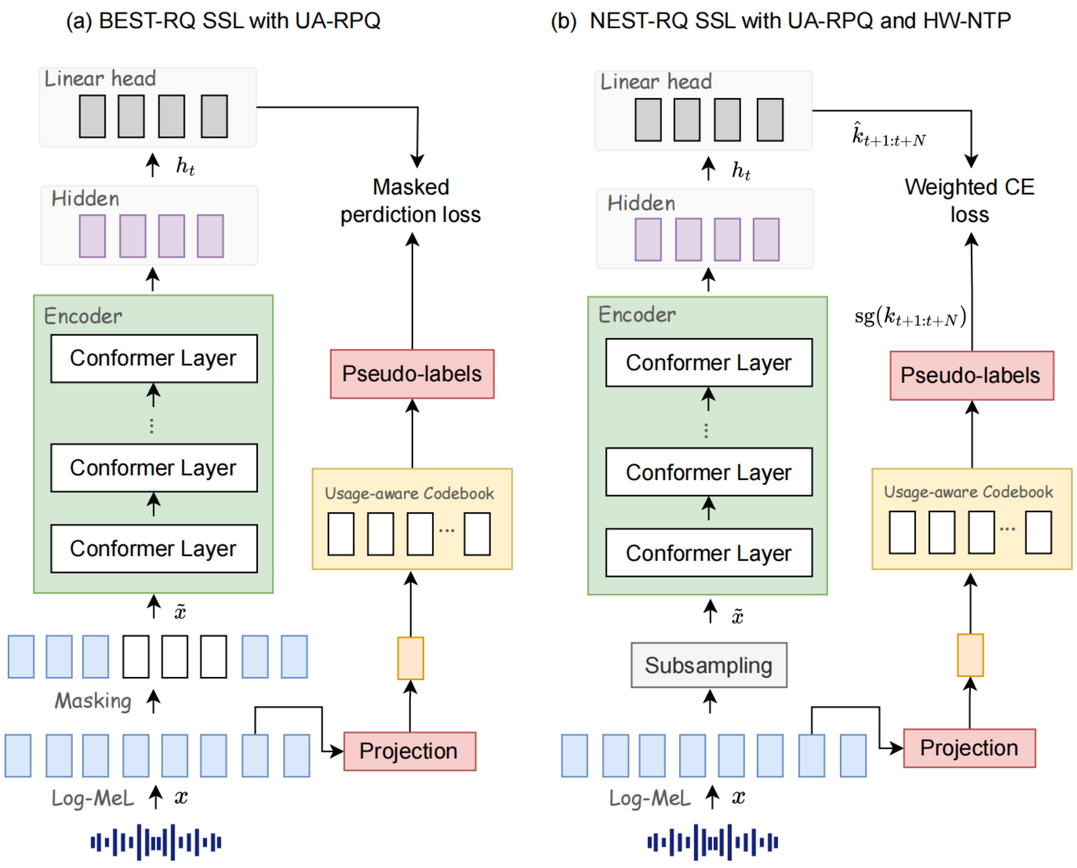


Figure 1. UA-RPQ supports BEST-RQ masked prediction and NEST-RQ multi-step prediction. HW-NTP reweights the future-token losses.

1. Usage-Aware Quantization

UA-RPQ keeps the random projection matrix fixed and adapts the codebook:

Usage tracking: Monitor code usage with an exponential moving average.

Adaptive detection: Identify underused codes relative to average usage.

Diversity regularization: Encourage balanced code assignments.

Anchor-based recycling: Reinitialize underused codes around active anchors.

The diversity loss updates only the codebook. Detached pseudo-labels supervise the encoder and prediction heads.

2. Horizon-Weighted Prediction

NEST-RQ predicts five future discrete speech tokens. HW-NTP assigns larger weights to nearer prediction horizons.

$$\widehat{w}_n = \frac{\gamma^{n-1}}{\sum_{m=1}^{N} \gamma^{m-1}}$$

We use N = 5 and $\gamma = 0.7$. Setting $\gamma = 1$ recovers uniform weighting.

EXPERIMENTAL SETUP

LibriSpeech: 960 h pre-training, 100 h fine-tuning.
WenetSpeech: 1,000 h pre-training, 100 h fine-tuning.
Encoder: 17-block Conformer, dimension 512, 8 heads.
Quantizer: 8,192 codes, code dimension 16.
Training: 100 pre-training epochs, CTC ASR fine-tuning.
Decoding: Beam search with an external 4-gram LM.

We transfer UA-RPQ hyperparameters and $\gamma = 0.7$ to WenetSpeech without dataset-specific tuning.

Training overhead: UA-RPQ adds about 3% per-step time and 35 MB peak memory in our experimental setup.

EXPERIMENTAL RESULTS

LibriSpeech: Non-streaming ASR

NEST-RQ configuration	test-clean WER (%)	test-other WER (%)
Baseline	6.68 ± 0.17	18.14 ± 0.27
+ UA-RPQ	5.95 ± 0.07	16.55 ± 0.14
+ HW-NTP	5.70 ± 0.08	14.65 ± 0.18
+ UA-RPQ + HW-NTP	**5.68 ± 0.10**	**13.98 ± 0.13**

Relative WER reduction vs NEST-RQ: 15.0% clean, 22.9% other.

WenetSpeech: Non-streaming ASR

NEST-RQ configuration	test-net CER (%)	test-meeting CER (%)
Baseline	19.87 ± 0.21	25.93 ± 0.34
+ UA-RPQ	17.55 ± 0.15	24.79 ± 0.28
+ HW-NTP	16.63 ± 0.14	23.32 ± 0.24
+ UA-RPQ + HW-NTP	**16.08 ± 0.11**	**22.03 ± 0.20**

Relative CER reduction vs NEST-RQ: 19.1% net, 15.0% meeting.

Mean ± standard deviation over three random seeds. Lower is better.

Streaming ASR: The combined method reduces LibriSpeech NEST-RQ WER from **15.05% to 13.83%** on test-clean and from **29.85% to 27.32%** on test-other.

UA-RPQ also improves the evaluated BEST-RQ, BiRQ, and RefineRQ models.

ANALYSIS AND ABLATIONS

Codebook Utilization

NEST-RQ configuration	Codebook entropy	Active codes (%)
Baseline	0.761	71.28
+ UA-RPQ	**0.864**	**85.37**

UA-RPQ increases entropy and active-code usage. Removing the adaptive threshold or disabling recycling worsens WER. Anchor-based recycling outperforms random reinitialization.

Prediction Horizons

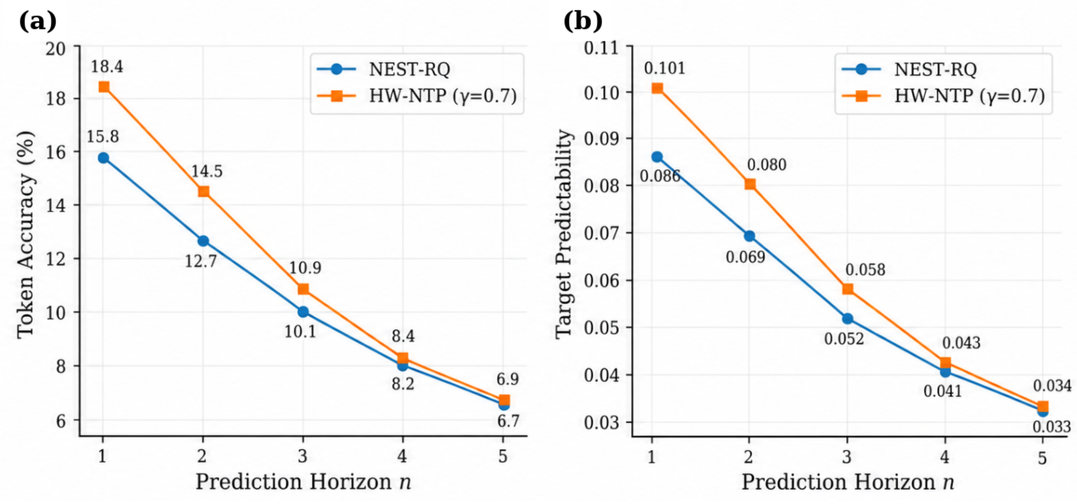


Figure 2. Top-1 token accuracy and target predictability across five future prediction horizons.

Both metrics decrease at longer horizons. HW-NTP improves both metrics, with larger gains at shorter horizons. Fixed $\gamma = 0.7$ gives the lowest WER among the evaluated weighting strategies.

CONCLUSIONS

UA-RPQ improves codebook utilization. HW-NTP accounts for future-token prediction difficulty. Their combination improves NEST-RQ ASR across both datasets and in non-streaming and streaming settings.